S24-A520U20-1

GCE A LEVEL

A520U20-1

eduqas
Part of WJEC

**PREDICTED PAPER – ADVANCED**

# ECONOMICS – A level component 2

## Exploring Economic Behaviour

2 hours 30 minutes

A520U201
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**ADDITIONAL MATERIALS**

A calculator.

A WJEC pink 16-page answer booklet.

**INSTRUCTIONS TO CANDIDATES**

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Answer **all** questions.

Write your answers in the separate answer booklet provided, following the instructions on the front of the answer booklet.

Use both sides of the paper. Write only within the white areas of the booklet.

Write the question number in the two boxes in the left-hand margin at the start of each answer, for example **1 1**.

Leave at least two line spaces between each answer.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

You are reminded of the necessity for good English and orderly presentation in your answers.

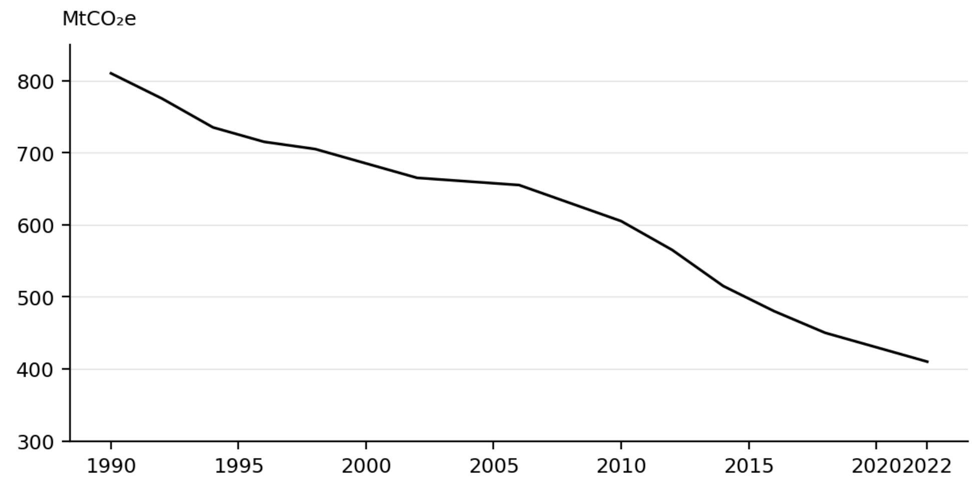
Answer **all** questions.

# Things can only get better?

The UK has positioned itself at the forefront of the global transition to a low-carbon economy. The Climate Change Act 2008 committed the country to legally binding carbon budgets, and in June 2019 the UK became the first major economy to legislate for Net Zero by 2050. Since 1990, UK territorial greenhouse-gas emissions have fallen by around 52% – further than in any other G7 country (**Figure 1**).

**Figure 1**

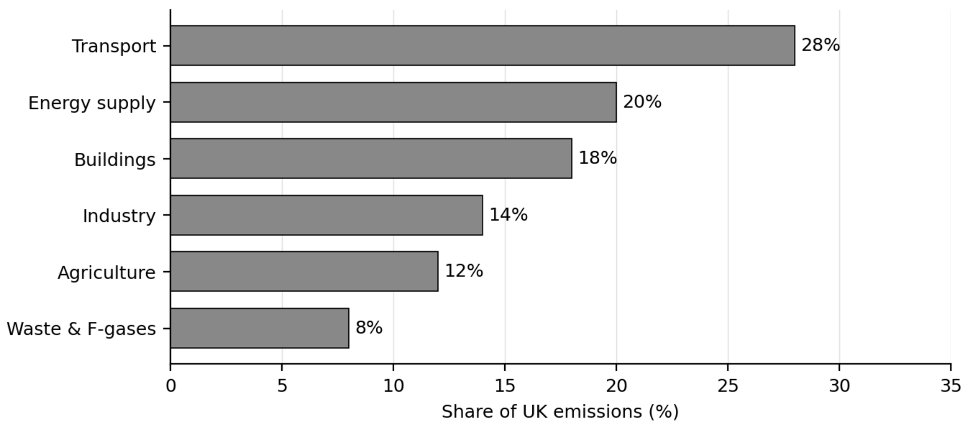
UK territorial greenhouse-gas emissions, 1990-2022




The bulk of the reduction so far has come from decarbonising electricity generation. Coal's share of UK electricity has fallen from 70% in 1990 to less than 2% in 2023. The next phase of the transition is harder: it requires decarbonising transport, heat in buildings, and the most energy-intensive parts of industry, sectors that still account for the majority of residual emissions (**Figure 2**).

**Figure 2**

UK emissions by sector, 2022 (% of total)

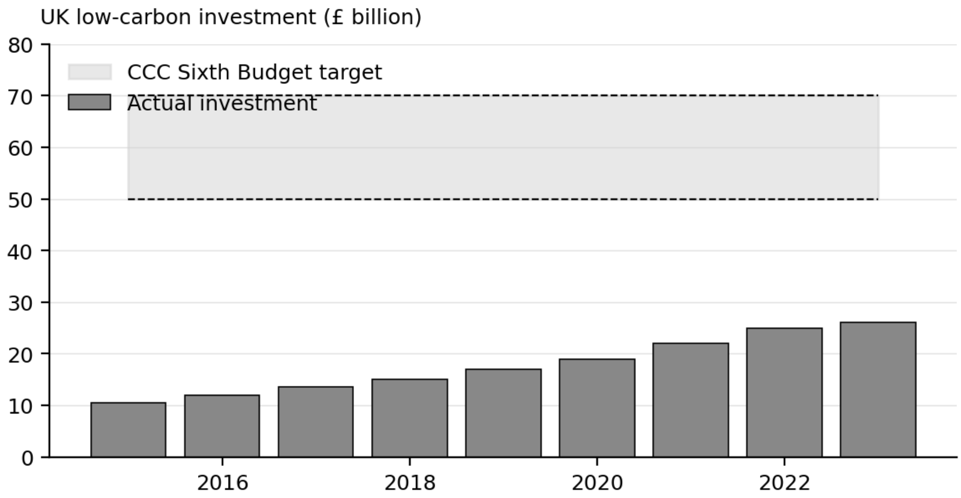



## The investment gap

The Climate Change Committee estimates that the UK needs £50-70 billion a year of low-carbon investment across the 2025-2030 period to meet the Sixth Carbon Budget. Actual investment is running well below that level (**Figure 3**).

**Figure 3**

UK low-carbon investment vs CCC target, 2015-2023

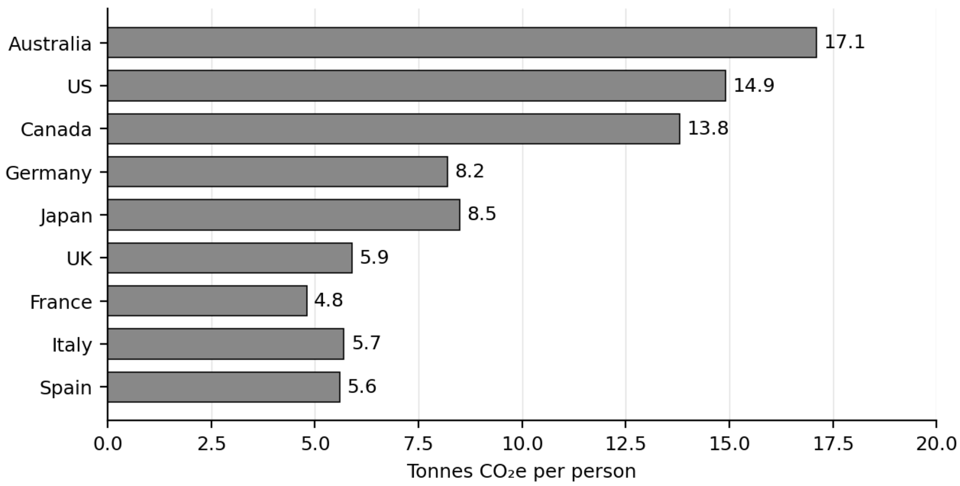



### International comparisons

The UK's emissions per capita have fallen faster than those of most other advanced economies. But comparisons also highlight the scale of the challenge: some major trading partners emit three times more per person (**Figure 4**).

**Figure 4**

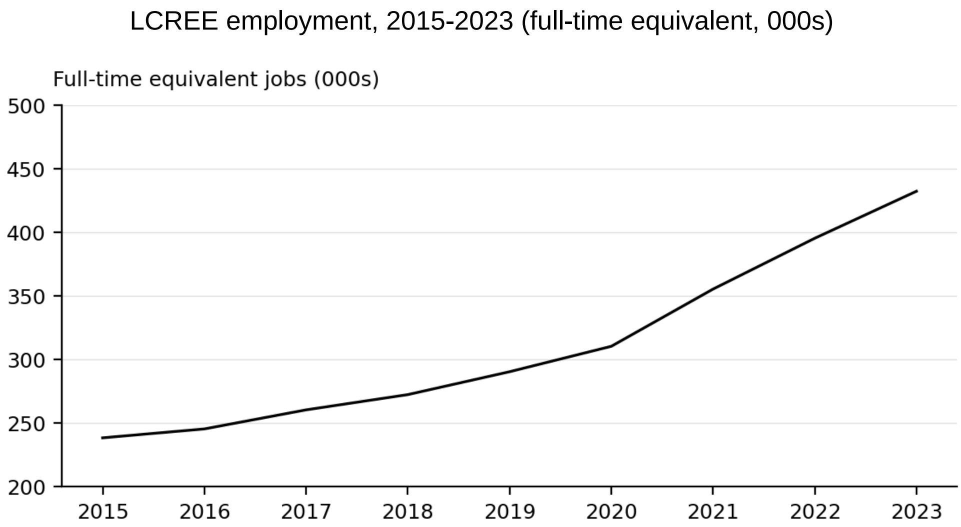
Emissions per capita, selected advanced economies, 2023




### The green economy

Employment in the low-carbon and renewable-energy economy (LCREE) has risen rapidly, reaching 432,000 jobs in 2023. The sector is growing faster than the UK economy overall (**Figure 5**).

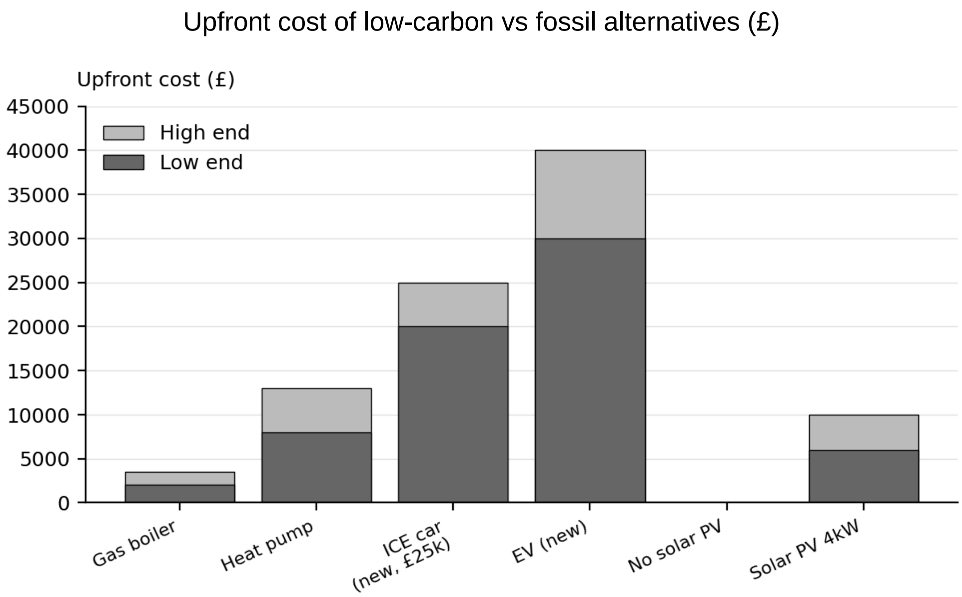
**Figure 5**




## The cost of going green

Low-carbon technologies typically cost more up-front than their fossil-fuel counterparts, even where running costs are lower (**Figure 6**). This creates a short-term affordability barrier for many households.

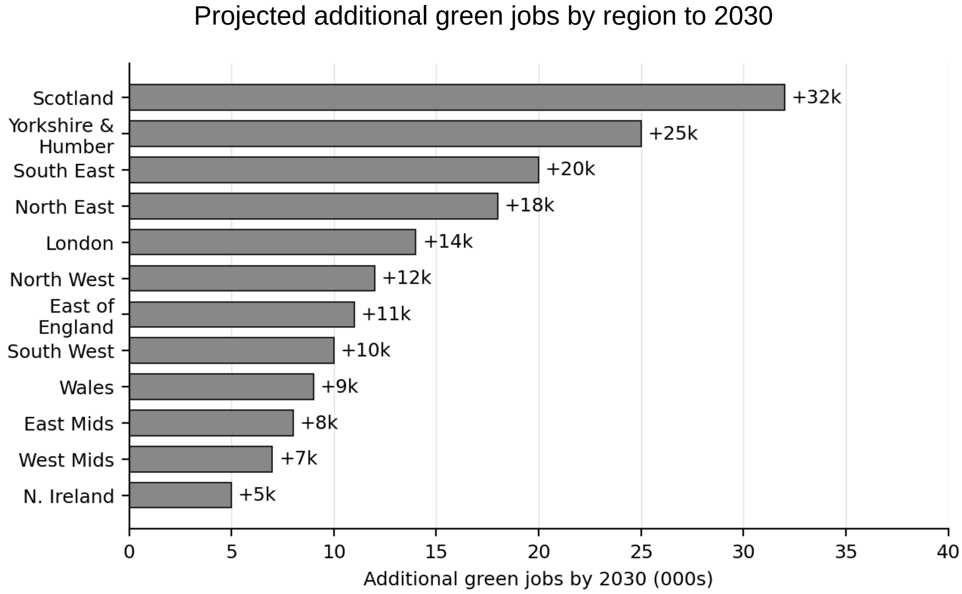
**Figure 6**




## Regional job creation

Green-economy jobs are not evenly distributed across the UK. Regions historically dependent on fossil-fuel extraction (Scotland, North East, Yorkshire & Humber) are projected to capture a disproportionate share of the offshore-wind, carbon-capture and hydrogen jobs created by 2030 (**Figure 7**).

**Figure 7**




## The UK's net-zero policy framework

The UK policy response combines carbon pricing, targeted subsidies, sectoral regulation and international measures (**Figure 8**).

| UK net-zero policy framework | |
|---|---|
| **Carbon pricing** | UK Emissions Trading Scheme (UK ETS): cap-and-trade covering power, industry and aviation. Cap declines by 2.2% per year.<br><br>Carbon Price Support tops up ETS prices for power generators to around £18/tonne. |
| **Low-carbon investment** | £20 billion programme for CCUS (Carbon Capture, Utilisation and Storage) industrial clusters by 2030.<br><br>Contracts for Difference auctions guaranteeing prices for offshore wind and other renewables. |
| **Demand-side subsidies** | Boiler Upgrade Scheme: £7,500 grant for heat pumps in owner-occupied homes.<br><br>Electric vehicle charging infrastructure investment (£1.6 billion by 2030). |
| **Regulation & standards** | Phase-out of new petrol and diesel cars from 2035 (extended from 2030 in Sept 2023).<br><br>New-build Future Homes Standard prohibits gas boilers in new homes from 2025. |
| **Border measures** | UK Carbon Border Adjustment Mechanism (CBAM) from 2027, covering imports of iron, steel, aluminium, cement, fertiliser and hydrogen. |

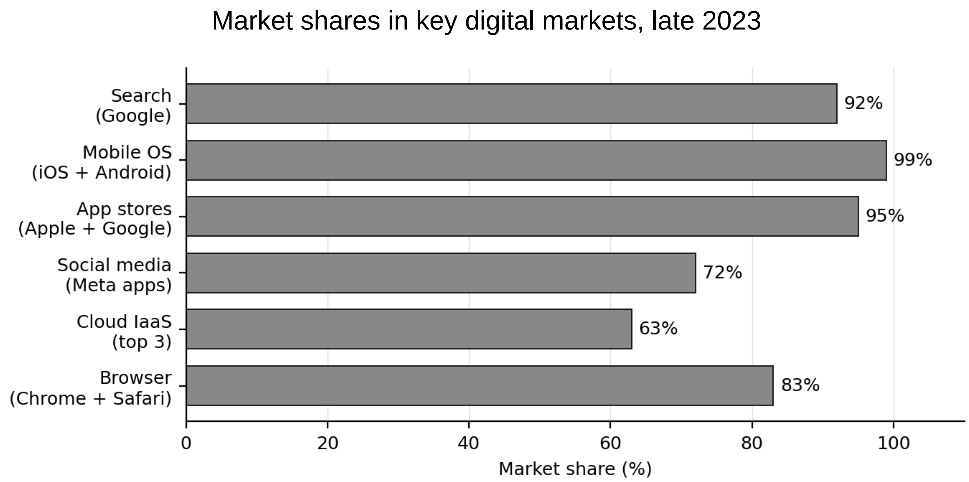
Sources: DESNZ, CCC, ONS, IEA.

| | | |
|---|---|---|
| **1 1** | Using the data, explain why UK greenhouse-gas emissions have fallen since 1990 (**Figure 1**). | [5] |
| **1 2** | Using the data, explain why the UK Government might want to accelerate investment in low-carbon technologies (**Figure 3**). | [5] |
| **1 3** | How effective is the UK's net-zero policy framework (**Figure 8**) likely to be in delivering the transition to a low-carbon economy? | [10] |
| **1 4** | Using the data, discuss the extent to which the distribution of green jobs across UK regions (**Figure 7**) is likely to reduce UK regional inequalities. | [10] |
| **1 5** | Using the data, discuss whether the higher upfront cost of low-carbon technologies (**Figure 6**) is likely to make the net-zero transition regressive. | [10] |

# The winner takes most

A small number of digital platforms now dominate the world's most strategically important markets. Google accounts for around 92% of global search; the iOS and Android operating systems together hold 99% of the global smartphone market; Apple and Google's app stores together control around 95% of the mobile-app distribution market; Meta's apps (Facebook, Instagram, WhatsApp, Messenger) are used by over 3.5 billion people (**Figure 1**). These firms combine the classical economic features of natural monopoly – very high fixed costs and near-zero marginal costs – with powerful network effects and massive data advantages.

**Figure 1**




The business models of these firms depend on scale. Apple and Google charge a standard 30% commission on app-store transactions, a rate that has attracted sustained criticism from developers and, since 2021, several regulatory interventions including the EU Digital Markets Act (DMA). Google pays Apple an estimated $18 billion a year to be the default search engine on iOS devices – a practice under investigation in the US antitrust case against Google that concluded in August 2024 with a finding that Google had illegally maintained a monopoly in search.

The top five firms (Alphabet, Apple, Meta, Amazon and Microsoft) spent around $225 billion on research and development in 2022 – roughly 14% of combined revenue, and more than the entire UK government R&D budget. They have also been the major force behind the commercialisation of generative AI from 2022 onwards.

The UK's Competition and Markets Authority (CMA) gained new powers under the Digital Markets, Competition and Consumers Act 2024 to impose "conduct requirements" on firms with Strategic Market Status (SMS). The first SMS designations are expected in 2025, covering mobile ecosystems and search. Conduct requirements may include mandatory interoperability, data portability, non-discrimination and price transparency – each of which would affect the abnormal profits earned by designated firms.

The data below (**Figure 2**) shows the cost and revenue schedule for PulseApp Ltd, a digital platform business. Demand is measured in thousands of monthly subscribers.

**Figure 2**

PulseApp Ltd: Cost and revenue schedule (per month)

| Price (£/ month) | Subscribers (000s) | Total Revenue | Marginal Revenue | Total Cost | Marginal Cost |
|---|---|---|---|---|---|
| **£18** | 0 | | | £500 | |
| **£16** | 100 | | | £520 | |
| **£14** | 200 | | | £560 | |
| **£12** | 300 | | | £620 | |
| **£10** | 400 | | | £720 | |
| **£8** | 500 | | | £860 | |
| **£6** | 600 | | | £1,060 | |
| **£4** | 700 | | | £1,340 | |
| **£2** | 800 | | | £1,720 | |

The table illustrates one of the most striking features of digital-platform economics: very high fixed costs (reflecting R&D, engineering and infrastructure spend) combined with very low marginal costs. Once the fixed cost of developing and running a platform has been sunk, the cost of serving an additional user is close to zero. This is a classic condition for natural monopoly, but unlike a traditional network utility, platform firms combine this cost structure with powerful demand-side network effects: each additional user makes the platform more valuable to every other user. The interaction of these two forces – supply-side economies of scale and demand-side network effects – is the core reason that digital markets tend towards winner-takes-most outcomes.

How should competition authorities respond? Supporters of the current market structure argue that consumers receive enormous benefits at zero price – search, email, social networking, maps – and that the rents earned by platforms fund innovation (generative AI being the latest example). Critics argue that the concentration of market power has become unprecedented, that innovation by rivals is suppressed through strategic acquisitions and entry barriers, and that classical remedies (breakup, structural separation) may be needed. Between the two positions lies a growing body of targeted regulation – the EU's DMA, the UK's SMS regime, the US antitrust litigation – that attempts to restore contestability without dismantling the firms.

Sources: Statcounter, Bloomberg, CMA, European Commission.

**2 1** Using the information on digital platforms, outline the characteristics of a market with substantial network effects and high fixed costs. [5]

**2 2** Using **Figure 2**, calculate the marginal revenue (MR) and marginal cost (MC) for each of the subscriber levels (0 to 800,000). Record the answers in your **pink** answer booklet. [7]

Using your answers explain at what level of output PulseApp Ltd will, in theory, maximise profits.

**2 3** Using a costs and revenue diagram, evaluate the effects on a dominant digital platform's abnormal profits of the imposition of regulation requiring it to offer interoperability with rival platforms. [10]

**2 4** Evaluate the view that competition authorities should break up dominant digital platforms in order to improve market outcomes. [7]

**2 5** With reference to the data, discuss the extent to which dominant digital platforms are harmful to economic welfare. [11]

**END OF PAPER**